Analysis of Traffic Accident Occurrence in Niigata Prefecture of Japan Using Open Data

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***Abstract*— In recent years, although the number of fatal traffic accidents tends to be decreasing, over 10 accidents per day still occur domestically on average. Primarily, physical improvement for automobiles, roads, and so on is important to eliminate traffic accidents. Also, we in parallel need radical traffic safety measures by analyzing traffic accident data and clarifying the traffic accident causes. On the other hand, utilization of public data owned by the government and municipalities, called Open Data, is positively promoted to solve various social issues. Therefore, we propose to find out novel features on the traffic accidents by analyzing both fundamental traffic accident data owned by each prefectural police department and Open Data in a multi-dimensional way, where Open Data include the data used for Geographic Information System. Additional findings regarding traffic accidents are unearthed by analyzing other Open Data such as municipal population data and meteorological data together. Examples using 45 months data of fatal traffic accidents in Niigata prefecture are presented.**

Keywords—open data, geographic information system, traffic accident, occurrence factor

I. INTRODUCTION

Vehicles are one of the most essential means of transportation for our daily life. On the other hand, they cause road accidents and the number of traffic accident victims is increasing worldwide. In Japan, 472,165 traffic accidents in 2018 took the life of 3,694 persons and injured 580,850 persons. Because of its aging population, the proportion of elderly people over the age of 65 in fatal accidents is increasing since 2011 and it has reached 54.6% in 2015.

In general, there are two ways to reduce the traffic accidents. One is physical and infrastructural measures. Such hardware side measures include attachment of an automatic breaking system and introduction of bumping structure to roads. The other is educational and legal measures. Such software side measures include traffic manner education and introduction of legal penal regulations. Toward implementing traffic accident countermeasures from either hardware or software side, it is important to identify accident-prone locations and to analyze spatial and temporal patterns of traffic accidents. Recently, because digital data for traffic accidents are easily available and visualizing methods become very popular, several studies of the analysis of traffic accidental pattern have been reported from each area of the world [1]-[4].

Shafabakhsh *et al.* [1] targeted at Mashhad, which is the second populous city in Iran. They used traffic accident records and applied a combination of geo-information technology and spatial-statistical analysis to bring out the influence of spatial factors in their formation. The areas were analyzed by statistical analysis methods and determined the eventfulness of the area based on standard deviation.

Prasannakumara *et al.* [2] targeted at a South Indian city and evaluated road accident hot spots. The patterns of localization and distribution of the hotspots were examined by making use of a geo-information technology. As the results, they brought out the influence of spatial and/or temporal factors in their formation.

Ghosh *et al.* [3] targeted at Dehradun, the capital of Uttaranchal in northern part of India. As the result of five years traffic accident analysis using Geographic Information System (GIS) technology, it was found that nearly 72% of accidents lead to fatal and grievous injuries. Further, it was observed that the number of accidents was the highest in May owing to tourist vehicles, because the number of medium type of vehicles was higher than any other type of vehicles. In addition, the occurrence of accidents was mostly concentrated between 2PM to 10PM. The study reveals that a proper traffic management is required for the city to check the growth of traffic accidents

Hirasawa and Asano [4] developed a traffic accident analysis system using a GIS for Hokkaido area in Japan. The traffic accident data for more than ten years could be visualized on a digital map and the information of road structure, road accessory facilities, and weather information were also added. By the developed system, accident data by road section and by specific condition could be extracted.

Those existing studies rely on a common analysis method. They collected location-tagged traffic accident data by a GIS system and just analyzed them by traditional statistical schemes.

In this paper, a novel analysis method is introduced for fatal accident data of Niigata prefecture in Japan. In the method, the location-tagged accident data are combined with the public data owned by the government and municipalities and analyzed from many viewpoints. These public data are called Open Data. As a result, traffic accident causalities that are unknown so far have been clarified.

II. FUNDAMENTAL ANALYSIS OF ACCIDENT DATA

The traffic accident data between January of 2013 and September of 2016 were offered by the Niigata Prefectural Police Department. The number of accidents is 367 for analysis. The data include the following items: a name of police station, accidental type, the number of people included in the accident, the accident date, the accident outbreak time, weather, address, latitude and longitude, the name of road, road shapes, existence of traffic lights, the accidental pattern, included vehicles (with the drivers' ages), accidental classification. Among these, the accidental type includes

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death accident, injury accident, and so on. In this research, only death accidents are treated, so that the accidental type is death accident for all of the data. The road shapes include intersection, near-intersection, straight, curb, and so on. The accidental pattern includes vehicle to vehicle (V2V), vehicle only (V), and vehicle to human (V2H). The accidental classification includes an aged person accident, a pedestrian accident, and a bicycle accident.

First, the accident outbreak number has been analyzed for each month. The analyzed result is shown in Fig. 1, where the horizontal axis indicates the month and the vertical axis indicates the accident outbreak number for the month. The features which Fig. 1 shows are that the accidental pattern of V2V relatively occurs between May and July and the pattern of H2V relatively occurs between December and January. One reason of the latter feature seems to depend on the fact that Niigata prefecture is famous of heavy snowfall in Japan.

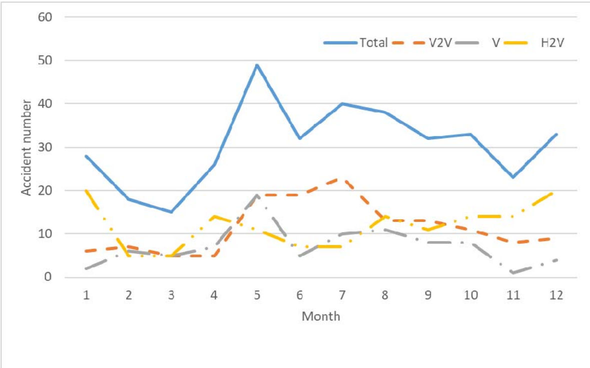

Fig. 1. Traffic accident data fundamental analysis: the accident outbreak number by month.

Next, the accident outbreak number has been analyzed for each hour. The analyzed result is shown in Fig. 2, where the horizontal axis indicates the time period when traffic accidents occurred and the vertical axis indicates the accident outbreak number for the hour. A feature which Fig. 2 shows is that the accidental pattern of V2V relatively occurs in the daytime. It is normal because people are more active and use cars more frequently in the daytime than in at night. One striking aspect in Fig. 2 is that the peak of the H2V pattern exists around 17:00 and the peak is relatively high.

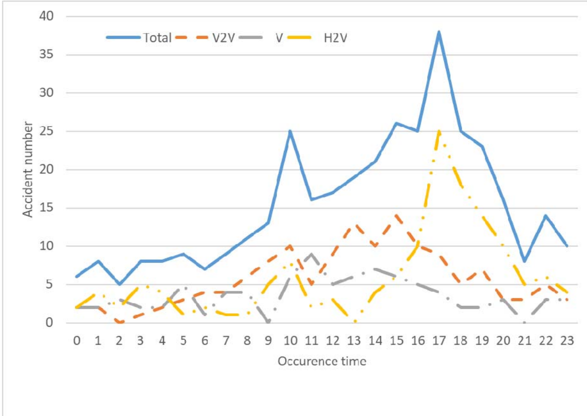

Fig. 2. Traffic accident data fundamental analysis: the accident outbreak number by hour.

III. ACCIDENT DATA ANALYSIS USING OPEN DATA

A. Combination with Meteorological Open Data

Japan Meteorological Agency measures precipitation amount, wind direction, wind velocity, temperature, day length, and snow depth at meteorological observatories located in various parts of Japan. These data are available via the Internet as Open Data. In addition, sunrise and sunset times in Niigata prefecture on the accident occurrence dates are utilized as Open Data. The analyzed results obtained from the accident data and these Open Data are presented in Fig. 3.

In Fig. 3, the horizontal axis indicates the day length rate and the vertical axis indicates the accident outbreak number. The day length rate is defined in this paper and it indicates the rate of day length time during one hour including the accident outbreak time. For example, if an accident occurs between 12:00 and 13:00 on one day and Meteorological Agency Open Day says the day length of the hour is 0.5, then the accident number is incremented for 0.5 of the day length rate. Meteorological Agency Open Day announces the day length as a value between “0” and 1. “0” does not, however, mean the night time. For the night time, Meteorological Agency Open Day indicates it as “NA” for the day length. In Fig. 3, “-1” of the day length rate corresponds to “NA”, that is the night.

Thus, the traffic accidents occurs the most frequently at the day length rate 0, which is not during the night but the day length is not available: namely just before the sunrise or just after the sunset. Especially, the number of H2V pattern accidents is the maximum and it corresponds to the fact that the peak of the H2V pattern exits around 17:00 in Fig. 2.

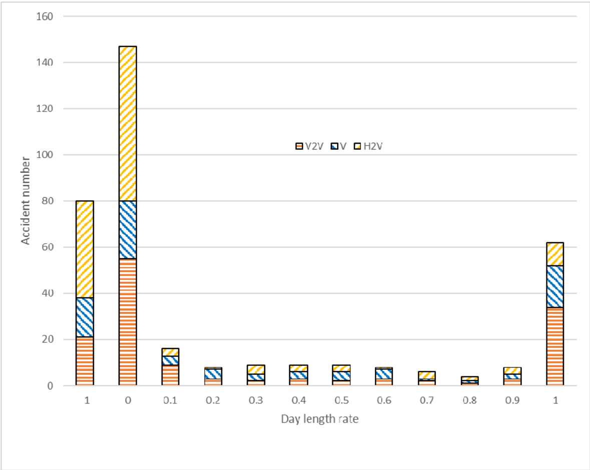

Fig. 3. Traffic accident data fundamental analysis: the accident outbreak number by month.

B. *Combination with GIS and Population Open Data*

Since digital maps published by the Geographical Survey Institute of Japan are available, the name of municipal government and the land usage of the accident location are easily revealed from the latitude and longitude information of the accident data. Moreover, since Ministry of Internal Affairs and Communications of Japan publishes the data of demographic and households based on Basic Resident Register in Japan every year, the relationship of population and accident occurrence for each municipality can be extracted accurately. A map of Niigata prefecture is shown in Fig.4

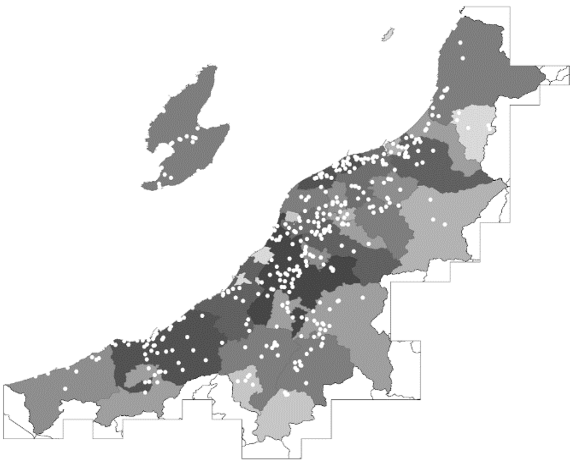

Fig. 4. Traffic accident location distribution and municipality division in Niigata prefecture. The darker the area color is, the more population the area has.

In Fig. 4, each municipality is colored a by 10-level gray scale. The darker it is, the more population it has. A small white circle indicates the accident occurrence location. It is found that traffic accidents tend to occur in the populated areas.

IV. Accident Data Analysis Using Image Data

It is possible to obtain images around the accident occurrence locations via the Internet owing to, for instance, Google Street View, without going to the location directly. Therefore, the circumstances around the intersection where a traffic accident occurred can be analyzed by using view images obtained through the Internet.

Although these are not actually Open Data, Explore world landmarks, discover natural wonders and step inside locations such as museums, arenas, restaurants and small businesses with Google Street View.

Focusing on the accidents occurred around an intersection in the day time, the images around the intersection are collected based on the latitude and longitude information of the accident location. A panoramic image is created from the collected images and the objects existing around the intersection are analyzed.

From the traffic accident data offered by the Niigata Prefectural Police Department, 43 panoramic images were created. It means 43 intersections were selected and they includes 33 four-way intersections and 10 three-way intersections. Examples of the panoramic images are shown in Figs. 5-8.

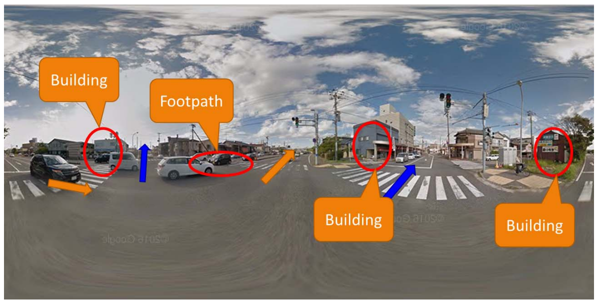

Fig. 5. An example of the panoramic images (four-way intersection).

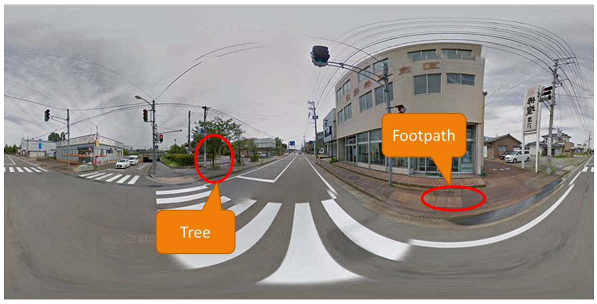

Fig. 6. An example of the panoramic images (three-way intersection).

Fig. 7. An example of the panoramic images (four-way intersection).

Fig. 8. An example of the panoramic images (four-way intersection).

Fig. 5 is an example of the four-way intersection, whose three corners have buildings and one corner has a footpath. Fig. 6 is a typical example of the three-way intersection. Fig. 7 is also a typical example of the four-way intersection and all of the corners have buildings that obstruct view. Fig. 7 presents an example of the four-way in a field.

Buildings, trees, and footpaths are the objects found at the corner of the intersection where an traffic accident occurred. Other objects found are traffic lights, signboards, posts (columns), fences, walls, roadways, parking places, farmlands, open fields. These objects were specified manually in this study.

As the relationship between accident occurrence and object existence, a case of traffic lights is described. Fig. 9 presents the analyzed result of the accident outbreak number and the existence of traffic lights. The horizontal axis indicates the accidental type: H2V, V2V, and V. The vertical axis indicates the total accident outbreak number for the intersections with and without traffic lights for each accidental type. It is found that traffic accidents apparently tend to occur at the intersection without traffic lights. The difference became more than triple in the case of the accidental pattern V2V.

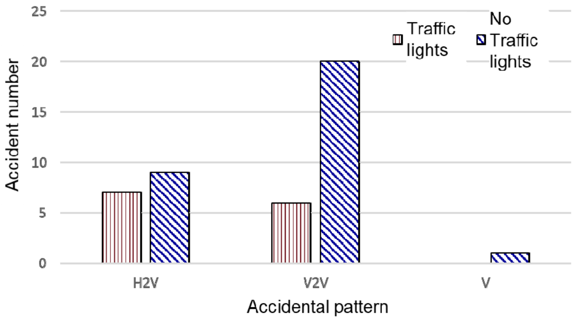

Fig. 9. The relationship between accident occurrence and traffic light existence for the accidental type: H2V, V2V, and V.

V. CONCLUSIONS

To make our life more safe and more secure, data analysis should be promoted, because a lot of variety and amount of data are easily available nowadays. In particular, utilization of the public data owned by the government and municipalities, so-called Open Data, are propelled not only for people's quality of life improvement but also for business promotion.

In this study, traffic accident data were analyzed by combining with Open Data and some novel findings obtained such analysis were presented. As Open Data, Japan Meteorological Agency measurement data, digital maps published by the Geographical Survey Institute of Japan, and population and household data from Ministry of Internal Affairs and Communications of Japan were used. The landscape images opened by private companies on the Internet are also utilized for the analysis, although they are not Open Data in a severe meaning.

Finally, it was clarified that a novel spatial and temporal analysis was shown by the combination of traffic accident data and Open Data.

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